The documentation and process conversion measures necessary to comply with this revision shall be completed by: 8 March 2020.

INCH-POUND

MIL-PRF-19500/279F
9 November 2019
SUPERSEDING
MIL-PRF-19500/279E
w/AMENDMENT 1
26 October 2010

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER,
TYPES 1N3644, 1N3645, 1N3646, AND 1N3647,
QUALITY LEVELS JAN AND JANTX

Inactive for new design after 7 June 1999. However, it is still valid for use as of 24 March 2004.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 0.1 ampere silicon rectifier diodes. Two levels of product assurance (JAN and JANTX) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outline. The device package for all device types is an axial leaded round body (similar to a DO−204AL) in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	V_{RWM}	V_R	I_O	I_O T_A = 100°C	$I_{(surge)}$ at V_R T_A = 25°C	T_J and T_{STG}	Altitude
	V	V	mA	mA	A	°C	Feet
1N3644	1,500	1,050	250	100	14	−65 to +175	40,000
1N3645	2,000	1,400	250	100	14	−65 to +175	40,000
1N3646	2,500	1,750	250	100	14	−65 to +175	40,000
1N3647	3,000	2,100	250	100	14	−65 to +175	40,000

1.4 Primary electrical characteristics. Not applicable to this specification sheet.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 5961

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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

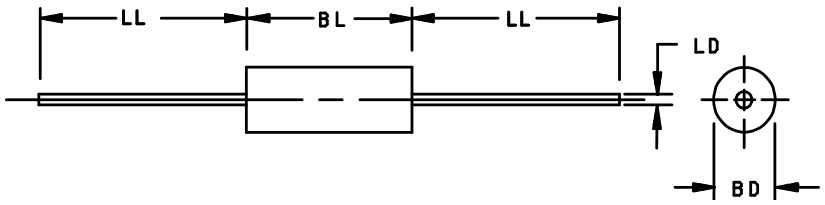
1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN" and "JANTX".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "3644", "3645", "3646", and "3647".

1.5.3 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.



Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.065	.110	1.65	2.79
BL	.190	.215	4.83	5.46
LD	.029	.033	0.74	0.84
LL	1.00	1.25	25.40	31.75

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for axial leaded package (similar to DO−204AL).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Encapsulating material(s). Encapsulating material(s) used shall be at the device manufacturer's option. Information concerning the material(s) shall be included with the qualification device design data.

3.4.3 Diode construction. All devices shall be metallurgically bonded, double plug construction in accordance with the requirements of category I, II, or III (see MIL-PRF-19500).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. The prefixes JAN and JANTX may be abbreviated as J and JX, respectively. The PIN may be reduced to J3644 or JX3644. Color coding shall not be acceptable for part marking. The polarity shall be indicated with a contrasting color band to denote the cathode end or the use of other techniques considered commercial practice.

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3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Process-conditioning, testing, and screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3 Process-conditioning, testing, and screening (quality level JANTX only). The procedure for process-conditioning, testing, and screening the JANTX types shall be in accordance with MIL-PRF-19500 and herein. Process-conditioning shall be conducted on 100 percent of the lot prior to submission of the lot to the tests specified in 4.3 through 4.4.3 herein.

4.3.1 High-temperature storage. All devices shall be stored for at least 48 hours at (T_A) = +175°C minimum.

4.3.2 Thermal shock (temperature cycling). All devices shall be subjected to thermal shock (temperature cycling) in accordance with method 1051 of MIL-STD-750, test condition C, except that T_{high} = +175°C and cycling duration shall be ten continuous cycles and exposure time at temperature extremes shall be 15 minutes minimum.

4.3.3 Pre burn-in tests. The parameters V_F and I_R shall be measured and the data recorded for all devices in the lot. All devices shall be handled or identified such that the delta end-points can be determined after the burn-in test. All devices which fail to meet these requirements shall be removed from the lot and the quantity removed shall be noted on the lot history.

4.3.4 Burn-in and life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees, nor less than 150 degrees.

4.3.5 Burn-in test. Power burn-in shall be in accordance with condition B of method 1038 of MIL-STD-750 for 164 hours (minimum). The following conditions shall apply:

a. T_A = 25°C, V_{RWM} = full rated (see 1.3).

b. I_O = 100 mA dc, f = 60 Hz.

c. Electrical measurements (end-points) shall be in accordance with screen 13 of 4.3.6, and table I, subgroup 2 herein.

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4.3.6 Screening. Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. Specified electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screening	JANTX level
2	48 hours, T_A = 175°C
3a	Required. T_{high} = 175°C minimum, ten continuous cycles, exposure time at temperature extremes shall be 15 minutes minimum.
9	Not required
10	Not required
11	I_R and V_F, see 4.3.3
12	See 4.3.5
13	Subgroup 2 of table I herein; ΔI_R = ±100 percent of initial value or 250 nA dc, whichever is greater; ΔV_F = ±10 percent of initial reading.

4.3.7 Post burn-in tests. The parameters V_F and I_R shall be retested (within 24 hours) after burn-in and the data recorded for all devices in the lot. The parameters measured shall not have changed during the burn-in test from the initial value by more than the specified amount as follows:

a. ΔV_F = 10 percent maximum.

b. ΔI_R = +250 nano-amperes or 100 percent maximum.

4.3.8 Burn-in test failure (screening). All devices that exceed the delta limits of screen 13 of 4.3.6, the post burn-in test of 4.3.7, and the maximum limits specified in table I subgroup 2 shall be removed from the inspection lot and the quantity removed shall be noted on the lot history. Where the quantity removed after burn-in exceeds 10 percent of the total inspection lot on burn-in test the lot may be resubmitted one time with a maximum PDA of 3 percent. For lots that have more than 20 percent failures or exceeds the 3 percent resubmission PDA on the initial burn-in, the entire lot shall be unacceptable as JANTX product.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB (JAN and JANTX) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1056	Condition A, except T_{low} = −40°C, ten cycles.
B2	1051	Condition C, 25 cycles.
B2	4066	Condition A, $I_{(surge)}$ = 14 A(pk), 10 surges of 8.3 ms each at 1 minute intervals, I_O = 0.1 A dc, V_{RWM} = rated V_{RWM} (see 1.3), T_A = +25°C.
B3	1026	I_O = 0.1 A, V_{RWM} = rated V_{RWM}, T_A = 100°C minimum, 340 hours duration, f = 50 to 60 Hz.
B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	1056	0°C to +100°C, 10 cycles.
C2	1051	−55°C to +175°C, 25 cycles.
C2	2036	Lead fatigue: Test condition E; weight 8 ounces (226.7 g).
C4		Not applicable.
C5		Not applicable.
C6	1027	I_O = 0.1 A, V_{RWM} = rated V_{RWM}, T_A = 100°C minimum, 1,000 hours duration, f = 50 - 60 Hz.
C7	1001	Barometric pressure, reduced: 40,000 feet; n = 22, c = 0. Monitor the rated V_R, I_R = 5 µA maximum.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward voltage	4011	Condition B; I_F = 250mA	V_F		5	V dc
Reverse current	4016	DC method	I_R			
1N3644		V_{RWM} = 1,500 V dc	I_R		5	µA dc
1N3645		V_{RWM} = 2,000 V dc	I_R		5	µA dc
1N3646		V_{RWM} = 2,500 V dc	I_R		5	µA dc
1N3647		V_{RWM} = 3,000 V dc	I_R		5	µA dc
1N3644		I_R = 50 µA	V_{BR}	1,800		V dc
1N3645			V_{BR}	2,400		V dc
1N3646			V_{BR}	3,000		V dc
1N3647			V_{BR}	3,600		V dc
Subgroup 3						
High temperature operation		T_A = +100°C; I_O = 100 mA dc				
Reverse current average	4046					
1N3644		V_R = 1,050 V ac	I_{RO}		100	µA dc
1N3645		V_R = 1,400 V ac	I_{RO}		100	µA dc
1N3646		V_R = 1,750 V ac	I_{RO}		100	µA dc
1N3647		V_R = 2,100 V ac	I_{RO}		100	µA dc

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

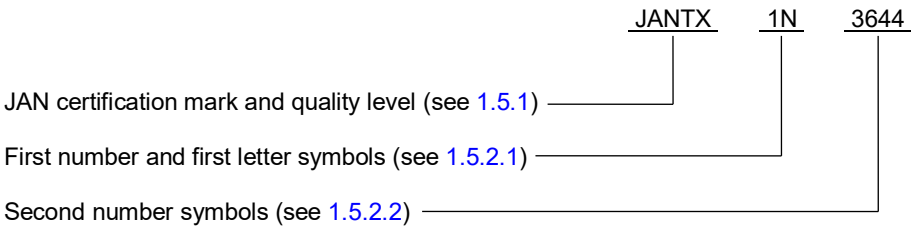
c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

e. Destructive physical analysis when requested.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction examples. The PINs for encapsulated devices and are constructed using the following forms.



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6.5 List of PINs. The following is a list of possible PINs available for encapsulated devices covered by this specification sheet.

Quality level "JAN"	Quality level "JANTX"
JAN1N3644	JANTX1N3644
JAN1N3645	JANTX1N3645
JAN1N3646	JANTX1N3646
JAN1N3647	JANTX1N3647

6.6 Substitution data. Device types 1N3643 and 1N3992 have been deleted from this specification sheet with the issuance of MIL-S-19500/279C (5 March 1971). Device type 1N3957 of MIL-PRF-19500/228 may be substituted for device type 1N3643. There is no direct replacement for device type 1N3992.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy − EC
DLA − CC

Preparing activity:
DLA − CC

Review activities:
Navy − OS, SH

(Project 5961−2019−109)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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